MENS MERCATUS

Katlego Mokoto

AI Engineer · AI Systems Architect · Full-Stack + .NET / C#

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Johannesburg, South Africa · Open to Johannesburg, Pretoria, Cape Town & Durban (on-site, hybrid or remote)

● AVAILABLE

AI engineer and developer — products, platforms, agentic systems, and ASP.NET backends.

Portfolio · mensmercatus.online — Portfolio proof: Cartel .NET robot (ASP.NET Core + Gemini), THINK (React 19 + Vite + TypeScript), Mens Mercatus Sound Deck, Forex Mercatus Codex, JobNet, Antigravity / local LLM stack. Walkthrough: mensmercatus.online/assets/video/walkthrough.mp4

ROLES

Systems architect

Full-stack engineer

.NET / C# developer

AI architect

STACK

Languages — Python, TypeScript, C#, JavaScript, SQL, Bash, Solidity

Frontend — React, React 19, Next.js, Vite, Flutter, Tailwind, Web Audio / DAW UIs

AI / ML — LLM orchestration, RAG, LangChain, Ollama, Agent Zero, Gemini (OpenAI-compatible API), prompt/tool agents, ChromaDB, local inferencing (Apple Silicon M4)

Backend — ASP.NET Core 8, .NET / C#, FastAPI, Node.js, Express, Django, WebSockets, Redis

Data — PostgreSQL, Pandas, NumPy, yfinance, OR-Tools

Cloud — Nebius, Docker, Linux, systemd, nginx, Cloudflare, CI/CD, AWS, Ubuntu VPS, edge / local-first AI

Audio — Web Audio API, mic capture, stem DSP, generative synthesis, live-voice WebSocket relay

Systems — P2P/BLE mesh, agentic workflows, Cursor IDE automation, per-IP rate limits

AVAILABILITY

Status — AVAILABLE

On-site, hybrid or remote · SA metros + global

FLAGSHIPS

Mercatus Assist / Cartel Robot LIVE

2026

ASP.NET Core 8 · Gemini · LIVE path via /api/cartel/ → VPS robot; DEMO fallback if backend unhealthy.

ASP.NET Core 8 minimal API (C#) that calls Gemini through an OpenAI-compatible chat/completions client so the provider can be swapped in config. Exponential backoff retries on 429/5xx. Runs as a systemd service on an Ubuntu VPS behind nginx; static site on DreamHost sits behind Cloudflare.

- Live Assist chat at mensmercatus.online/apps/mercatus-assist/ with labelled DEMO fallback when the robot is down.
- Same-origin PHP proxy → VPS; architecture notes at mensmercatus.online/apps/net-workspace/.

`C# · ASP.NET Core 8 · Gemini · systemd · nginx · Cloudflare`

THINK — Hybrid Social & AI LIVE

2026

React 19 + Vite + TypeScript · Public demo with per-IP daily caps on chat, image, video, and live-voice.

Multimodal Gemini workspace (chat, live voice, Net Sites, Net Code, Netzar Protocol, Net Learn). Front end is React 19 + Vite + TypeScript; server-side Node proxy calls Gemini and relays live-voice over WebSocket, with per-IP rate limits.

- Live at mensmercatus.online/think/.
- 60s portfolio walkthrough: mensmercatus.online/assets/video/walkthrough.mp4.

`React 19 · Vite · TypeScript · Gemini proxy · WebSocket live-voice`

SoundX / Sound Deck DEMO

2026

Browser DAW / AI music · Honest demo — sound stays in the browser, not a hosted session.

Original browser DAW with AI Producer (order-taking studio assistant), demo stem split, 16-step beats, mic/vocal pipelines, VST slot framework, and in-house text-to-music / stem / continuation generation.

- Generate, demo play, mic, and AI Producer on mensmercatus.online apps hub.
- Gold/black/white Mens Mercatus product UI; no third-party music-site dependency.

`Web Audio · Agents · DSP · TypeScript/JS`

JobNet DEMO

2026

Career automation · Draft-only — live read-only shortlist demo; never applies or sends mail.

Weekday pipeline: noon multi-board scrape; CV tailor + AI Hawk draft fills with human confirm before submit; Gmail reply watch + SMS/email notify.

- Noon scrape across top AI employers + early-stage trackers.
- 3 AM CV tailor + draft fills; human-in-the-loop gates before any submit.

`Automation · Agents · Gmail · macOS shell`

Forex Mercatus Codex DEMO

2026

Python desk :8790 · Simulated paper research only — not a brokerage.

~87MB Python + Highcharts desk for FX workflows; Chart, Multi, Whales, Flow, Bot. Hosted on VPS as live paper research.

- Inventory and load-path optimization for faster cold start and clearer static serving.
- Market data fused with interactive analysis desks.

`Python · Highcharts · yfinance · Paper only`

SELECTED WORK

Project Antigravity

Mar 2026 – Present

AI Systems Architect & Founder

- Sovereign, decentralized AI OS for autonomous task execution and specialized agentic workflows.
- Terminal-first AI tooling on M4 with Ollama; Nebius for scalable training/deploy of custom models; Agent Zero + Cursor orchestration.

`Ollama · Agent Zero · Nebius · LLM Ops`

Mens Mercatus Ecosystem

Ongoing · mensmercatus.online

Full-stack, .NET & AI systems

- Telegram AI trading assistant: RAG + live financial feeds for XAUUSD / cyclical models (SMT, Gann frameworks) with institutional-order context.
- Brand, UI/UX, and portfolio architecture including the Cartel ASP.NET Assist robot and the THINK multimodal workspace.

`Python · C# / ASP.NET · LangChain · ChromaDB · React / Next.js`

NARRATIVE

- Hands-on builder of agentic products: copilots that execute studio/job/market actions with explicit user confirmation gates.
- Comfortable in Python, TypeScript, and C# / ASP.NET Core; ships Gemini-backed APIs on a VPS (systemd + nginx) and React 19 fronts on the portfolio.
- Background in fleet/logistics optimization (OR-Tools) and diesel/mechanical operations — strong systems thinking under real constraints.

Portfolio: mensmercatus.online · Homelake / Randfontein · Open to remote AI eng roles